Homework 3

Q Question:

Write a C program to manage a simple user login and grade management system with the following requirements:

1. User Authentication:
  - The program should accept a **User ID (UID)** (integer) and a **Password** (string) from the user.
  - These inputs should be passed to a function `password_match(uid, password)` that:
    - Reads a file named `passwords.txt`. Each line in the file contains a UID and its corresponding password in the format:
    ```
    **UID, password**
    ```
    - Validates if the provided credentials are correct.
    - Returns `**true**` if valid; otherwise, returns `**false**`.

2. If Credentials Are Valid:
  - Access an ****index file**** named `**index.txt**`, which contains user details in the format:
    ```
    **UID,Name,Course1:Grade1,Course2:Grade2,...,CourseN:GradeN**
    ```
  - Extract the user details based on the given UID.
  - Use a ****Record structure**** to store the user's name, a list of up to 4 courses, and their corresponding grades (as integers).

3. Display a Menu:
  After successful login, display a menu with the following options:
  - **1. View Details**:
    Display the user's name, courses, and grades.
  - **2. Add New Course**:
    Allow the user to add a new course and its grade. Update the `**index.txt**` file with this new entry. Ensure the program doesn't exceed 4 courses for a user.
  - **3. Get Grade for a Specific Course**:
    Prompt the user for a course name and display the corresponding grade. If the course doesn't exist, display an appropriate message.
  - **4. Exit**:
    Exit the program.

4. **Error Handling**:
  - If the login credentials are invalid, display an error message and terminate the program.
  - Ensure all file operations are safe and provide error messages if files are missing or corrupted.

Expected Files:

1. **passwords.txt** (for authentication):
   ```
   101,password1
   102,password2
   103,password3
   ```

2. **index.txt** (for user details):
   ```
   101,John Doe,Math:85,Science:90
   102,Jane Smith,History:75,English:80
   103,Tom Brown,Physics:70,Chemistry:65
   ```

Example Run:

Input:
```
Enter UID: 101
Enter Password: password1
```

Output (if credentials are valid):
```
Login successful!

Menu:
1. View Details
2. Add New Course
3. Get Grade for a Specific Course
4. Exit
Enter your choice: 1
```

Output for Option 1:
```
User Details:
Name: John Doe
```

Courses and Grades:
  Math: 85
  Science: 90
```

Input for Option 2:
```
Enter your choice: 2
Enter the new course name: Physics
Enter the grade for Physics: 88
```

Output:
```
New course added successfully!
```

Updated `index.txt`:
```
101,John Doe,Math:85,Science:90,Physics:88
102,Jane Smith,History:75,English:80
103,Tom Brown,Physics:70,Chemistry:65
```

Input for Option 3:
```
Enter your choice: 3
Enter the course name: Math
```

Output:
```
Grade for Math: 85
```

Exit:
```
Enter your choice: 4
Exiting...
```

Note : Based on this Home assignment end sem question will asked like following example.
```

consider the last assignment now write a function to compute the GPA of student based on their grades, which easly integrate in the last assignment.

This type of question may asked in the end semester exam so complete the home assignment without failure.